

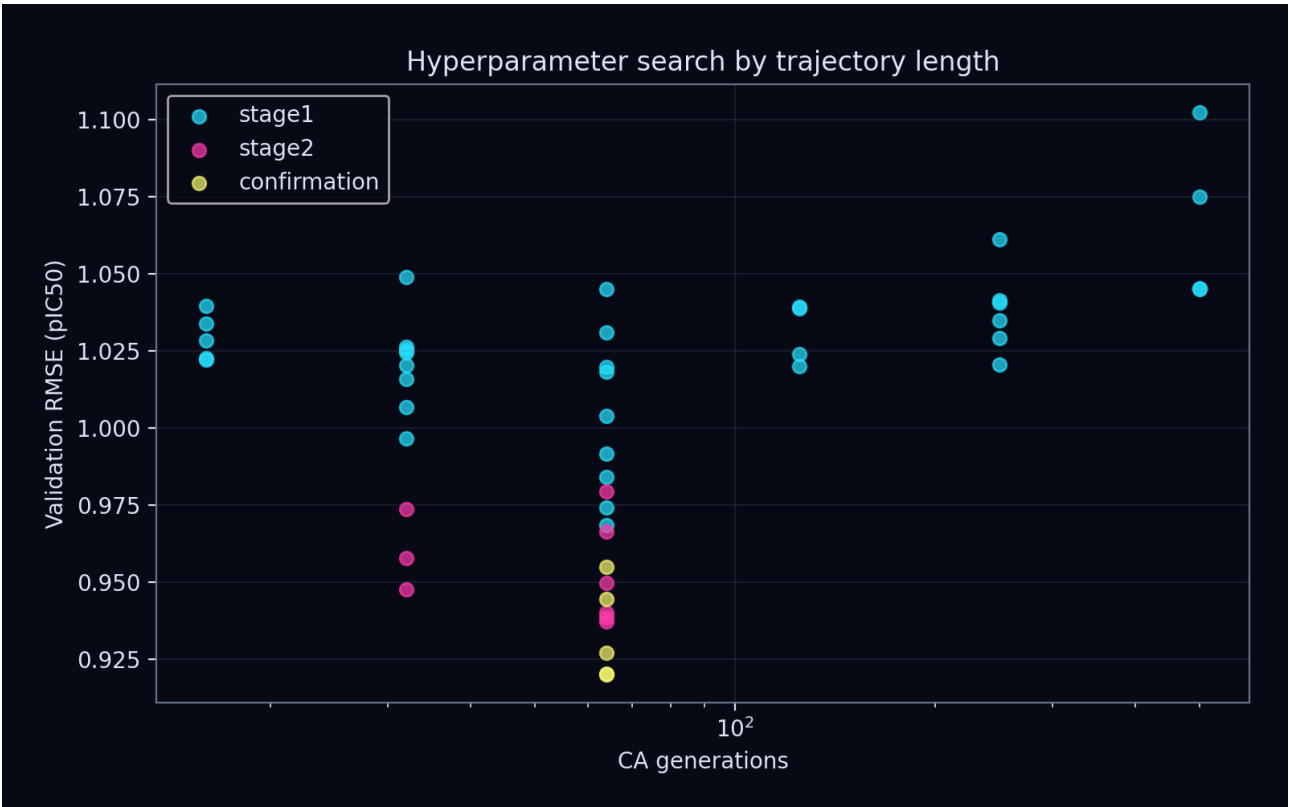
# Strange Matter Engine

## Production study: kuramoto\_sakaguchi

Direct-inhibition pIC50 | grouped validation | blinded set excluded

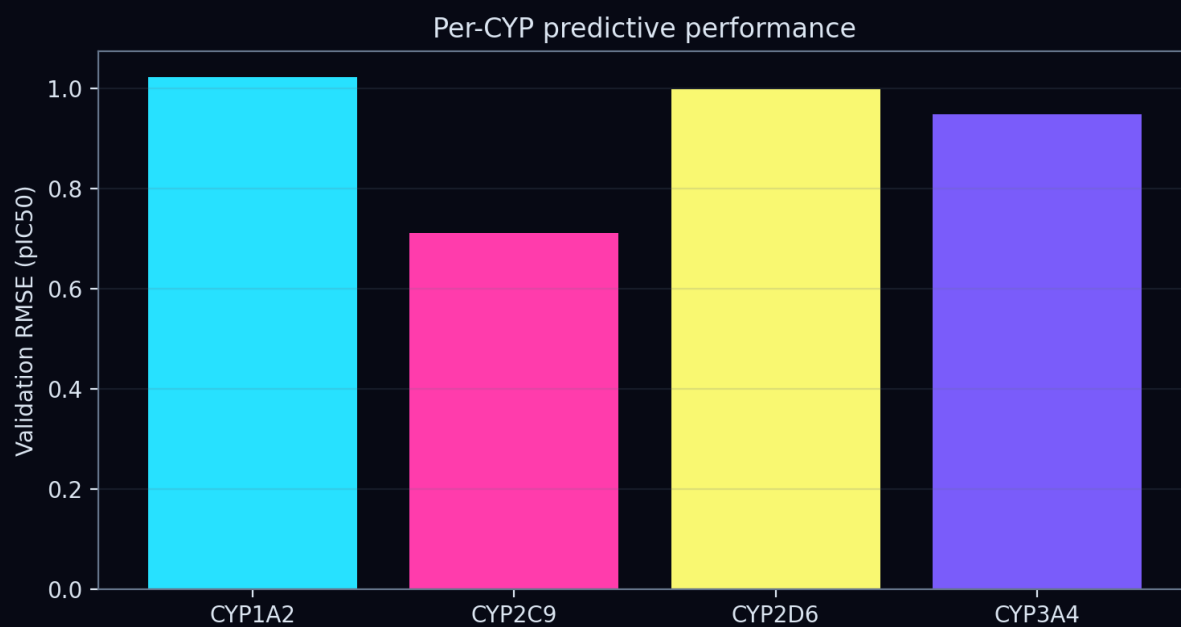
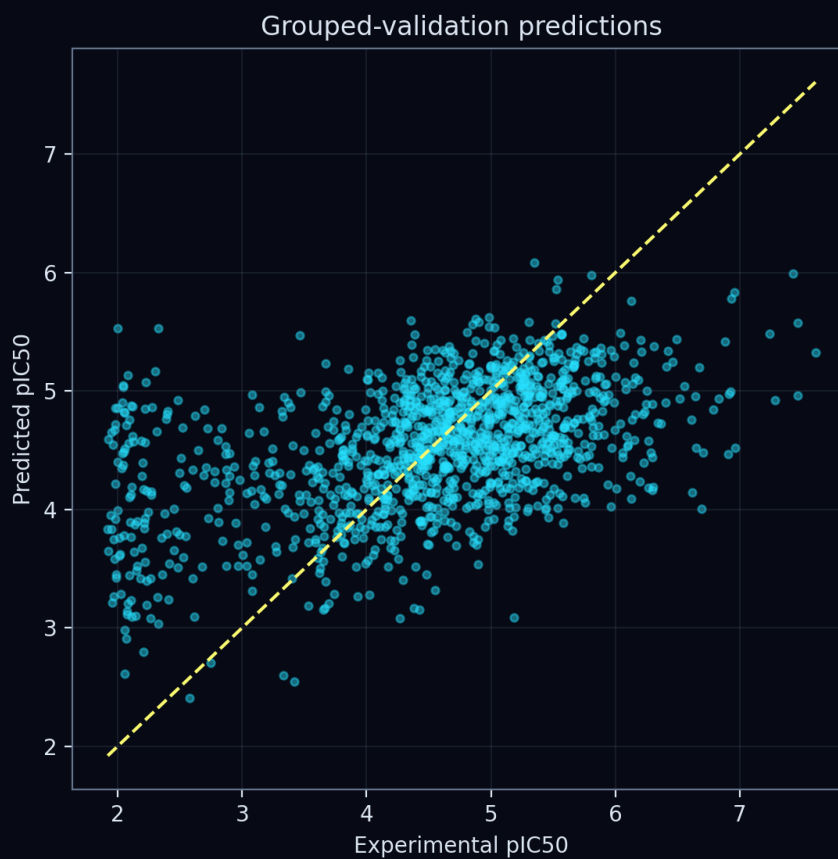
| Quantity               | Result       |
|------------------------|--------------|
| Validation RMSE        | 0.9374 pIC50 |
| Fit RMSE               | 0.8866 pIC50 |
| Generations            | 64           |
| Dynamical channels     | 8            |
| CA learning rate       | 0.001        |
| Ridge penalty          | 0.1          |
| CA L2                  | 1e-05        |
| Gradient clip          | 1.0          |
| Confirmation mean RMSE | inf          |
| Confirmation seed SD   | nan          |

Enhanced CA controls: update scale 0.25, initial-state scale 1.0, support fraction 0.75, bond temperature 1.0. Rule dynamics: A=0.5, B=0.5, C=0.5, D=0.2.



## Predictive validation

Model selection used grouped-validation RMSE only. Dynamical-interest scores did not influence promotion.



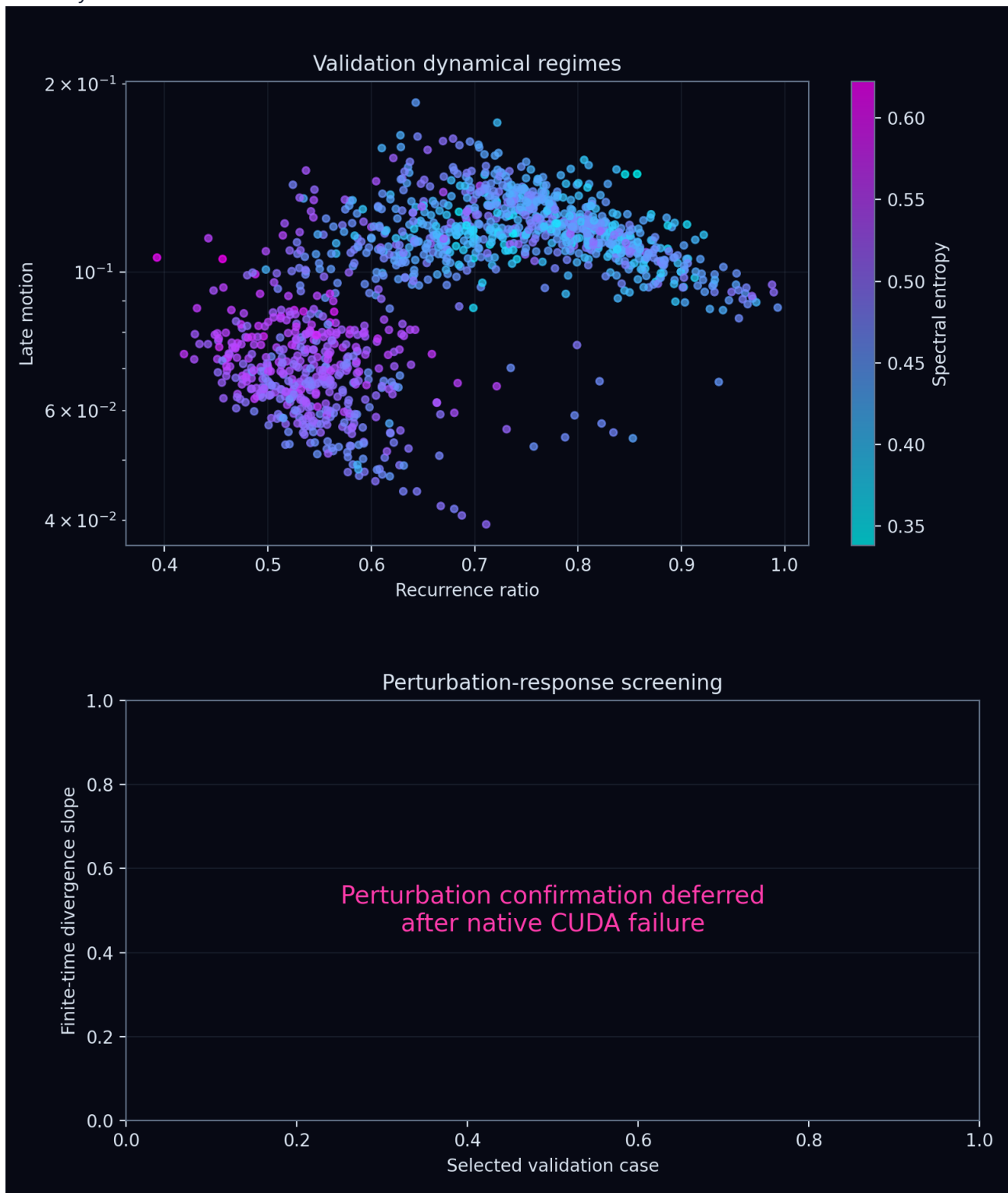
## Optimization

Ridge coefficients and the unpenalized intercept were solved from permitted fitting fingerprints. Query error differentiated through the solve into the graph CA; Adam updated only CA parameters. Bond-conditioned messages, cosine learning-rate decay, and multitime trajectory statistics were shared across all five rules.



## Emergent dynamics

The validation trajectories were screened for convergence, recurrence, persistent motion, spectral complexity, and perturbation sensitivity. Labels ending in candidate require longer, renormalized and seed-repeated confirmation before any attractor or chaos claim.



## Scientific limitations

This report describes one transition-rule study under the declared grouped split. Hyperparameter selection and early stopping used labelled training data only. The blinded challenge set was not loaded or predicted. Finite-time positive slopes are screening signals rather than proof of a strange attractor or sustained chaos.